Research Statement

Dayanara M. Diaz Vargas

October 2026

My research uses causal inference to measure the effects of public policy on economic opportunity and access to public services in historically marginalized communities. Its goal is to help governments design programs that reach the people they intend to serve. I focus on vulnerable populations, including workers in declining regions, patients navigating healthcare access barriers, and patients in medically underserved areas, in contexts where market failures and institutional constraints disproportionately affect those with limited resources or bargaining power. An ideal and understudied setting for this is Puerto Rico, where 37.3 percent of residents lived in poverty in 2024, about twice the rate of the poorest state (U.S. Census Bureau, 2025). Over the past five decades, the island has experienced the withdrawal of a long-standing federal tax credit that supported manufacturing, a shift to private management of publicly financed healthcare, and a mass migration of physicians following Hurricane Maria. I draw on these events to examine how changes in government-sponsored programs and healthcare supply affect employment and health. These projects complement research on policy adoption and incremental reform by identifying the distribution of their effects across places and demographic groups. To identify these effects, I rely on staggered difference-in-differences for rollouts across counties, synthetic difference-in-differences for shocks to a single economy, and instrumental variables paired with a structural model for location choices shaped by earlier migration. Within health economics and labor economics, my work contributes to the literature on place-based policies focused on tax incentives, healthcare privatization, and physician location choice. Going forward, I will extend this agenda to other types of place-based policies that affect labor markets and public services, and I will collaborate with policymakers and community organizations to turn my findings into actionable policy. Additionally, through teaching and mentorship, I intend to encourage students (particularly those from underrepresented backgrounds) to research policies affecting their own communities.

In my job market paper, I study the effect of health care privatization on patient's health. Evidence on privatization in health care systems usually isolates one institutional change at a time, such as who owns a facility, how providers are paid, or who administers coverage. Puerto Rico implemented a policy that changed the whole system all together, providing a rare setting to study this phenomenon. Between 1994 and 2000, the government sold or leased its hospitals and clinics, replaced government salaried providers with private capitated contracts, and moved low-income residents into an income-based insurance program run by private organizations, while care remained publicly financed. Because this health reform reached counties at different times, I compare counties that had been reformed with counties not yet reformed, which provides a counterfactual for what would have happened without the change for the same population. The reform increased all-cause mortality by 3.7 percent of its pre-reform mean, relative to the counterfactual. It also reduced the number of ambulatory health care establishments and the share of births with early prenatal care, linking the rise in mortality to lower provider capacity and lower healthcare access. To explain these results, I adapt the framework of Hart, Shleifer, and Vishny (1997), in which private managers keep what they save and therefore have strong incentives to cut costs. Their framework states that since healthcare is a credence good, its quality is hard to specify in a contract, and thus some of those savings can come from cutting care itself. This hypothesis supports the losses in capacity, access, and health that I find. In doing so, my paper provides evidence that patient health and provider capacity matter when evaluating broad reorganizations of publicly financed care.

My second paper explores labor markets and studies the withdrawal of a long-standing place-based corporate tax incentive. From 1976, Section 936 of the Internal Revenue Code exempted U.S. corporations from federal tax on their Puerto Rico income, and by 1992 the manufacturing establishments that claimed it employed

42.2 percent of the island's manufacturing workers. Congress phased the credit out between 1996 and 2006. Because Puerto Rico is the single treated unit of observation in my state-level panel dataset, I use synthetic difference-in-differences, which builds a counterfactual from a weighted combination of U.S. states. Using data through 2010, I estimate that the phase-out lowered manufacturing employment by about 15 percent and the working-age population by about 5 percent, while the estimate for total employment is close to zero and imprecise, separating losses within manufacturing from the island's overall employment adjustment. In an analysis through 2000, manufacturing losses were about twice as large in high-poverty counties as in the rest. These results suggest that losing corporate tax incentives can scar local industries for years, most deeply in the poorest communities, and the decline in the working-age population points to migration as one way the local workforce adjusted.

My ongoing work on physician location choice connects these strands by asking how the allocation of skilled workers affects access to essential services. It asks how much an additional physician improves health and what incentives are needed to attract physicians to underserved communities, a question that can more broadly speak to the labor market frictions on locally delivered (nontradable) services. Both questions are difficult because physicians choose where to practice, so places with more physicians may differ from places with fewer for reasons that also affect health. In joint work with Dr. Jori Barash, we use physician relocation after Hurricane Maria in 2017, with an instrument based on pre-existing Puerto Rican diaspora settlement patterns, to predict which mainland communities received physicians. We will estimate effects on preventable hospitalizations and examine whether benefits differ for Hispanic patients; ethnic concordance between patients and physicians is one potential explanation for such a difference, which we treat as a hypothesis. A structural model of physicians' location choices will then translate these estimates into recruitment counterfactuals, estimating the compensation needed to change where physicians practice. Together, these pieces could help recruitment programs weigh the health returns to placing physicians against the cost of doing so, and a possible extension would measure what Puerto Rico's own counties lost when these physicians left.

My next project extends my study of public services from health care to public safety. Beginning in 1993, Puerto Rico's government sent police and National Guard troops into public housing to fight drug-related crime, and by the end of 1995 they had occupied 72 housing projects. Campaigns like this one concentrate state enforcement in the poorest neighborhoods, yet whether they save lives is still debated, since they may reduce violence, raise lethal violence, or push it into nearby communities. Because the occupations reached housing projects at different times, I will use a staggered difference-in-differences design that compares places already occupied with places not yet occupied, treating lethal violence as a health outcome, using county vital-statistics records I have already assembled, and accounting for other changes on the island in the same years. Comparing effects in poorer and better-off communities, and in areas next to those with occupations, will show how the campaign's consequences were distributed and how far they reached. The project will contribute to the evaluation of public-safety interventions and carry my agenda's central question of the effects of public services available in disadvantaged places.

A second strand of my future research examines whether place-based policies intended to attract investment, encourage work, and retain skilled professionals improve economic opportunities and access to care in disadvantaged communities. Under Puerto Rico's Act 22 of 2012, new residents received tax breaks on investment income unavailable to longstanding residents. I will examine whether the benefits of these tax incentives extend to the residents they exclude through local employment opportunities, using decree holders' settlement patterns to measure county exposure. A complementary project studies incentives that support low-income workers directly. For the federally supported 2021 expansion of Puerto Rico's earned income tax credit, I will examine employment responses using differences across groups in how the reform changed credit generosity. Extending my work on physician mobility, I will also study the retention incentives introduced

under Act 14 of 2017, which reduced physicians' taxes in exchange for practicing on the island and providing community service. I will examine the effect this physician retention policy had on CHIP, Medicaid, and Medicare beneficiaries health. As I refine these designs, my aim is to understand how incentives directed at investors and workers translate into employment opportunities and healthcare for disadvantaged residents.

Across these projects, my research examines who benefits and who bears the costs when governments change the economic support and public services available in economically depressed areas. The underlying economic questions extend well beyond Puerto Rico, including how poor regional industries adjust when government subsidies are reduced, how public services fare under private management, and how skilled professionals are spatially allocated though different tax incentives. Ultimately, this agenda aims to show how public policies affect employment, health, and the distribution of their benefits and costs, which is the evidence governments need to design policies that are both efficient and equitable.